Design-Weighted LCA: Segments, Prediction, and Handoff

Institutional trust in Mexico: LAPOP AmericasBarometer 2023

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1 Purpose

This script is the fast, interpretive half of the pipeline. It reads the fitted objects the modeling script saved to `{cfg$out_dir}`, drafts (or reads) the segment labels, presents the measurement model with design-based intervals alongside the unweighted `poLCA` benchmark and its validation verdict, predicts segment membership back to every respondent who answered enough items, profiles the segments on the configured demographics (one page each), and writes the analyst handoff: a labelled `.sav` plus CSVs. Throughout, a latent class is called a **segment**, a word-choice preference; the statistical object is unchanged.

2 What the segment labeler is asked

Before any label is trusted, here is the instrument verbatim: the system prompt (the persona) and one real user prompt, built for segment 1 from this survey's own wording and probabilities. Three properties to note while reading: the prompt carries only ONE segment (isolated calls, an empirical design: joint prompts confused near-neighbor segments); every probability is shown against the item's actual response labels; and rule 1 fences the survey context to referent resolution, so nothing outside the shown probabilities may be attributed to a segment. This instrument, including the segment terminology, is certified by `llm_label_obedience_experiment.R`; any edit to it obliges a re-run.

```r
meta_preview <- item_meta(survey_dat, cfg$items, cfg$na_codes, questions)
cat("=========== SYSTEM PROMPT (persona) ===========\n")
```

```
=========== SYSTEM PROMPT (persona) ===========
```

```r
cat(lca_persona(), "\n\n")
```

```
You are a senior survey methodologist who reads latent class analysis (LCA) measurement models
```

```r
cat("=========== USER PROMPT (segment 1) ===========\n")
```

```
=========== USER PROMPT (segment 1) ===========
```

```r
cat(prompt_class_label(fit_w, 1, meta_preview, cfg$items, cfg$survey_context), "\n")
```

```
SURVEY CONTEXT
These items come from the 2023 AmericasBarometer survey of Mexico, conducted by the LAPOP Lab a

The battery analyzed here measures trust in national institutions: respondents rate, for each i
ONE SEGMENT FROM A LATENT CLASS ANALYSIS (LCA) MEASUREMENT MODEL
SEGMENT 1 (estimated prevalence 33%):
justice_system "Confianza en el sistema de justicia"
P(Nada)=0.00, P(2)=0.04, P(3)=0.15, P(4)=0.46, P(5)=0.29, P(6)=0.05, P(Mucho)=0.02
electoral_tribunal "Confianza en el Tribunal Supremo Electoral"
P(Nada)=0.03, P(2)=0.05, P(3)=0.12, P(4)=0.27, P(5)=0.29, P(6)=0.17, P(Mucho)=0.08
armed_forces "Confianza en las fuerzas armadas"
P(Nada)=0.02, P(2)=0.02, P(3)=0.07, P(4)=0.17, P(5)=0.28, P(6)=0.24, P(Mucho)=0.20
legislature "Confianza en la legislatura"
P(Nada)=0.02, P(2)=0.03, P(3)=0.10, P(4)=0.38, P(5)=0.30, P(6)=0.13, P(Mucho)=0.04
public_ministry "Confianza en el Ministerio Público"
P(Nada)=0.01, P(2)=0.02, P(3)=0.10, P(4)=0.37, P(5)=0.35, P(6)=0.11, P(Mucho)=0.05
police "Confianza en la policía nacional"
P(Nada)=0.08, P(2)=0.17, P(3)=0.29, P(4)=0.26, P(5)=0.14, P(6)=0.04, P(Mucho)=0.02
auditor "Confianza en la Auditoría Superior de la Federación"
P(Nada)=0.01, P(2)=0.02, P(3)=0.13, P(4)=0.41, P(5)=0.30, P(6)=0.10, P(Mucho)=0.02
political_parties "Confianza en los partidos políticos"
P(Nada)=0.10, P(2)=0.19, P(3)=0.28, P(4)=0.29, P(5)=0.10, P(6)=0.03, P(Mucho)=0.00
president "Confianza en el presidente"
P(Nada)=0.04, P(2)=0.06, P(3)=0.07, P(4)=0.17, P(5)=0.19, P(6)=0.21, P(Mucho)=0.26
supreme_court "Confianza en el tribunal supremo"
P(Nada)=0.00, P(2)=0.04, P(3)=0.11, P(4)=0.31, P(5)=0.37, P(6)=0.13, P(Mucho)=0.04
municipality "Confianza en su municipalidad"
P(Nada)=0.01, P(2)=0.04, P(3)=0.14, P(4)=0.37, P(5)=0.34, P(6)=0.08, P(Mucho)=0.02
media "Confianza en los medios de comunicación"
P(Nada)=0.05, P(2)=0.10, P(3)=0.21, P(4)=0.27, P(5)=0.25, P(6)=0.08, P(Mucho)=0.04
elections "Confianza en las elecciones"
P(Nada)=0.02, P(2)=0.05, P(3)=0.21, P(4)=0.30, P(5)=0.26, P(6)=0.13, P(Mucho)=0.02

TASK
Read this single segment and return: a short DRAFT label (2 to 5 words) for an analyst to refi

RULES:
1. Use only the response probabilities and item wording shown. The survey
   context only clarifies what the items refer to; attribute nothing to
   the segment that the probabilities do not show.
```

```
2. Anchor every statement to the high-probability answers of this segment.
3. If the profile is diffuse (no clear high-probability answers), say so.
4. Return only valid JSON: no prose before or after, no markdown fences.
JSON (one object): {"label": "...", "description": "..."}
```

3 Naming the segments

One rule: `segment_labels.csv` in `{cfg$out_dir}` is used when it exists (validated against K); otherwise the LLM drafts once, one call per segment, and writes it, with any harmonizer edit recorded in `Label_draft`. **Editing that file is taking over the naming**; the labels below are drafts to verify against the profile heatmap.

```r
# get_class_labels() implements the one rule above (survey_lca_source.R,
# Section 4). The cache file freezes the LLM output for reproducibility.
# cfg$data, NOT dat_prepared: recode_consecutive() strips attributes, so a
# recoded copy would silently cost the prompts all wording and labels.
segment_labels <- get_class_labels(fit_w, survey_dat, cfg$items, cfg,
                                   questions = set_names(dictionary$question,
                                                         dictionary$item))

knitr::kable(segment_labels,
             caption = "Segment labels and descriptions in use (segment_labels.csv in cfg$out_
```

Table 1: Segment labels and descriptions in use (segment_labels.csv in cfg$out_dir). Verify each against its response profile before quoting; edit the file to take over naming.

K	Label	Description
1	Moderate Institutional Trust	This segment shows a preference for moderate to high trust (scores 4-7) across most institutions, with the strongest confidence in the president and armed forces. Trust is notably lower for political parties and the national police.
2	Very High Institutional Trust	This segment is characterized by very high trust in the president (82%), public ministry (74%), and armed forces (72%), with a general tendency to select 'Mucho' across nearly all national institutions.
3	High Institutional Trust	This segment expresses high levels of trust across all measured institutions, with the strongest probabilities concentrated on scores of 5, 6, or 7, particularly for the president, armed forces, and the supreme court.
4	Generalized Institutional Distrust	This segment is characterized by very high probabilities of 'Nada' (not at all) across almost all institutions, most notably political parties (0.85) and the national police (0.76). The only notable exceptions are the president and armed forces, which show more distributed responses and higher probabilities of trust compared to other institutions.

K	Label	Description
5	Presidential and Military Trust	This segment is characterized by high trust in the president and armed forces, contrasted with very low trust in political parties and the national police.

```
# Display labels for every downstream table and figure: short, wrapped, fixed
# class order so factor levels are stable everywhere.
# Three display tiers, because one label format cannot serve every surface:
#   lab_of    wrapped factor, for FIGURE legends and facets (newlines are fine there)
#   lab_line  single line "Ck: Label", for SMALL tables (a newline inside a kable
#             cell breaks Typst layout)
#   lab_short "Ck" only, for LARGE tables; the class-label table above is the legend
lab_lev   <- stringr::str_wrap(paste0("S", segment_labels$K, ": ", segment_labels$Label), 22)
lab_of    <- function(k) factor(lab_lev[k], levels = lab_lev)
lab_line  <- function(k) paste0("S", k, ": ", segment_labels$Label[k])
lab_short <- function(k) paste0("S", k)
```

4 The measurement model: inclusion probabilities, benchmark, and validation

One table carries the design-weighted segment prevalences with their JKn intervals, the unweighted `poLCA` prevalences, and the gap. The printed validation line is the trust check: at unit weights our pseudo-EM and `poLCA` maximize the same likelihood, so their prevalences must match to optimization tolerance before the weighted output deserves belief.

```
max_diff <- max(abs(fit_unw$pi - pl_aligned$pi))
cat(sprintf("VALIDATION: max |pi difference|, unit-weight pseudo-EM vs poLCA = %.5f -> %s\n",
            max_diff,
            ifelse(max_diff < 0.005, "MATCH (implementations agree)",
                   "MISMATCH - investigate before trusting weighted results")))
```

```
VALIDATION: max |pi difference|, unit-weight pseudo-EM vs poLCA = 0.00494 -> MATCH (implementa
```

```
# ci_tbl columns (from the modeling script): kind, item, category, class,
# estimate, se, lo, hi. Renamed here to short names for the display code.
pi_ci <- ci_tbl |> dplyr::filter(kind == "pi") |>
  transmute(K = class, est = estimate, lo, hi)
incl_tbl <- pi_ci |>
  mutate(Segment = lab_line(K),
         `Weighted pi [95% CI]` = sprintf("%.3f [%.3f, %.3f]", est, lo, hi),
         `poLCA (unweighted)`   = sprintf("%.3f", pl_aligned$pi[K]),
         Gap = sprintf("%+.3f", est - pl_aligned$pi[K])) |>
```

```
  dplyr::select(Segment:Gap)
knitr::kable(incl_tbl, align = "lrrr",
             caption = "Segment inclusion probabilities: design-weighted estimates with JKn in
```

Table 2: Segment inclusion probabilities: design-weighted estimates with JKn intervals, the unweighted poLCA benchmark, and the weighting gap. The validation line above certifies the implementations agree at unit weights.

Segment	Weighted pi [95% CI]	poLCA (unweighted)	Gap
S1: Moderate Institutional Trust	0.331 [0.292, 0.371]	0.336	-0.005
S2: Very High Institutional Trust	0.111 [0.091, 0.132]	0.111	+0.000
S3: High Institutional Trust	0.209 [0.171, 0.246]	0.207	+0.001
S4: Generalized Institutional Distrust	0.107 [0.084, 0.129]	0.107	+0.000
S5: Presidential and Military Trust	0.242 [0.202, 0.282]	0.239	+0.004

```
pi_ci |>
  mutate(Segment = lab_of(K)) |>
  ggplot(aes(Segment, est)) +
  geom_pointrange(aes(ymin = lo, ymax = hi), linewidth = 0.5) +
  geom_point(aes(y = pl_aligned$pi[K]), shape = 5, size = 2.6) +
  scale_y_continuous(limits = c(0, NA)) +
  labs(x = NULL, y = "Inclusion probability") +
  theme_lca() +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))
```

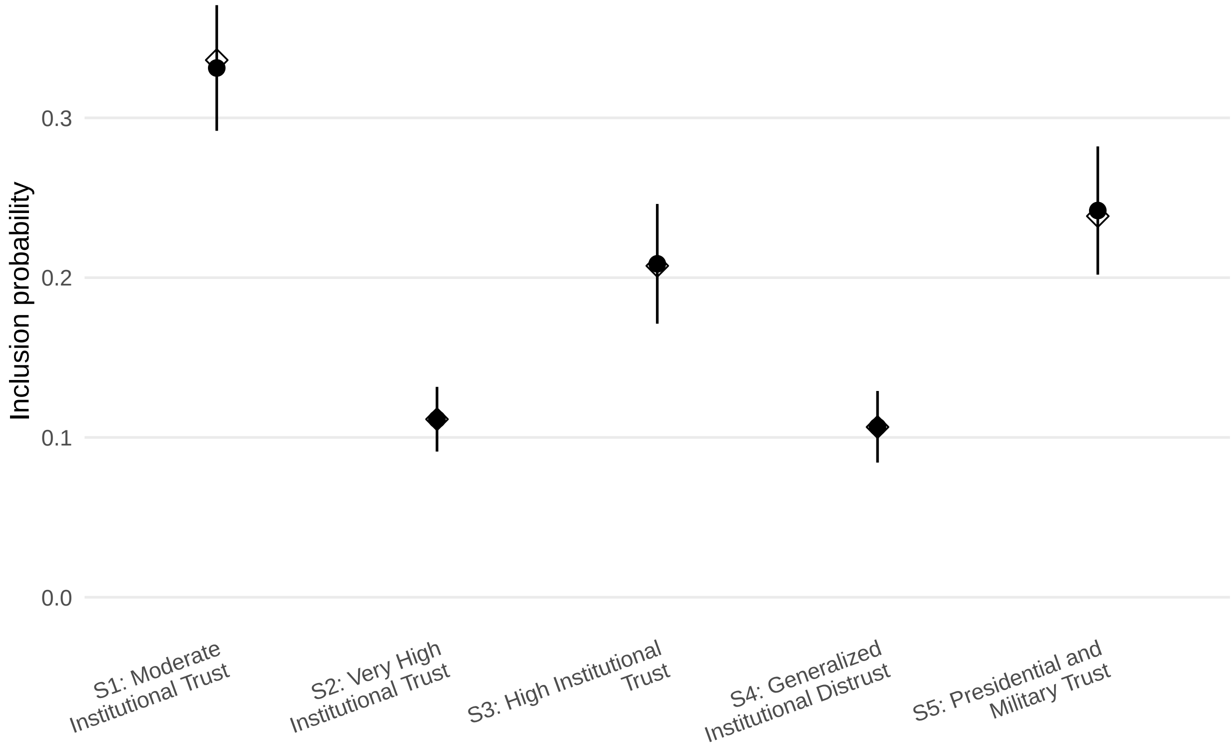


Figure 1: Weighted segment prevalences with 95% JKn intervals (points and bars) against the unweighted poLCA benchmark (open diamonds). Gaps are what the design corrects.

5 Segment profiles: conditional response probabilities

```
ci_tbl |>
  dplyr::filter(kind == "rho") |>
  mutate(Segment = lab_of(class),
         item = factor(item, levels = cfg$items),
         est = estimate) |>
  ggplot(aes(factor(category), item, fill = est)) +
  geom_tile() +
  geom_text(aes(label = sprintf("%.2f", est)), size = 2.1) +
  scale_fill_viridis_c(name = "P", limits = c(0, 1)) +
  facet_wrap(~ Segment, ncol = min(3, K_star)) +
  labs(x = "Response category", y = NULL) +
  theme_lca() +
  theme(axis.text.y = element_text(size = 7))
```

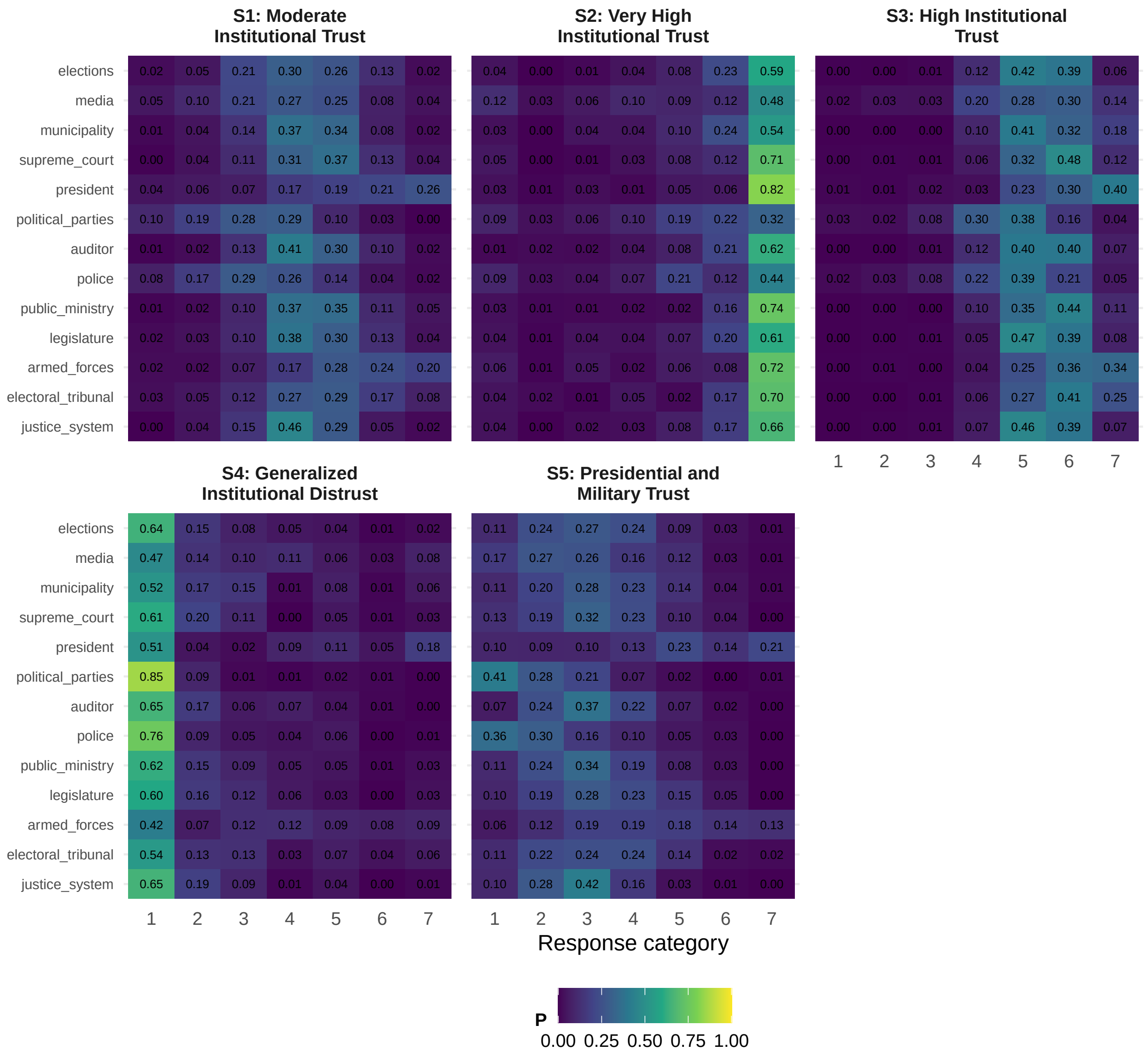


Figure 2: P(response | segment) for every item. Panel headers carry the drafted segment labels; verify each label against its panel before quoting it.

6 Predicting segment membership

Prediction runs on the FULL preserved frame: local independence lets a missing item drop out of the within-segment product, so partial responders are predictable, with `cfg$min_items_predict` as the evidence floor (below it, no prediction rather than a guess). Items must be coded exactly as fitted; the config script guarantees that here.

```
pred <- predict_segments(survey_dat_original, fit_w, cfg$items,
                         min_items = cfg$min_items_predict)
survey_scored <- bind_cols(survey_dat_original, pred)
```

```
cat(sprintf(
  "Predicted %d complete and %d incomplete cases of %d total (floor: %d items); mean max poste
  sum(!is.na(pred$segment) & survey_scored$keep),
  sum(!is.na(pred$segment) & !survey_scored$keep),
  nrow(survey_scored), cfg$min_items_predict,
  mean(pred$max_posterior, na.rm = TRUE)))
```

```
Predicted 1430 complete and 188 incomplete cases of 1622 total (floor: 7 items); mean max poste
```

7 Segment composition by demographics

Design-based segment prevalence within each demographic level: the design-weighted mean posterior in that subpopulation, with JKn intervals on the logit scale, the standard domain estimate. One demographic per page, table above plot, so nothing crowds.

```
# The segment-membership target is the soft posterior attached in the setup
# chunk; post_cols are the K posterior columns in segment order. des and rep_des
# (the design and its JKn replicates) were also built in setup.
post_cols <- paste0("post_class", seq_len(K_star))

# Fixed layout for the flattened statistic: every (demographic, level, class) cell,
# levels in factor order, class varying fastest within a level. dom_meta names and
# orders the estimate vector so nothing is recovered by string parsing later.
dom_meta <- map_dfr(cfg$aux, function(v)
  expand_grid(plevel = levels(dat_prepared[[v]]), class = seq_len(K_star)) |>
    mutate(pvar = v)) |>
  dplyr::select(pvar, plevel, class) |>
  mutate(idx = row_number())

# The statistic survey::withReplicates evaluates on each replicate. For each
# demographic it forms a level-indicator matrix, then the weighted mean of each
# class posterior within each level: crossprod(M, w * P) / crossprod(M, w). The
# result is flattened to ONE named numeric vector in dom_meta's order (level-major,
# class-minor), because withReplicates needs a vector to build the covariance.
domain_theta <- function(w_rep, data) {
  P  <- as.matrix(data[, post_cols, drop = FALSE])      # n x K, fixed across replicates
  wP <- w_rep * P                                       # n x K, weighted posteriors
  flat <- map(cfg$aux, function(v) {
    f   <- factor(data[[v]], levels = levels(dat_prepared[[v]]))
    M   <- model.matrix(~ f - 1)                        # n x L, one column per level
    num <- crossprod(M, wP)                             # L x K: weighted class sums per level
    den <- as.numeric(crossprod(M, w_rep))              # L: weighted total per level
    as.vector(t(num / den))                             # level-major, class-minor
  })
  unlist(flat, use.names = FALSE)
```

```
}

dom_rep <- withReplicates(rep_des, domain_theta)        # design-based covariance of every cel
dom_est <- domain_theta(weights(des), dat_prepared)     # full-sample point estimates
dom_V   <- vcov(dom_rep)                                # full covariance, for the contrasts be
dom_se  <- sqrt(diag(dom_V))
stopifnot(length(dom_est) == nrow(dom_meta))
```

```
# Per-level class prevalence with a design-based interval. The interval is formed
# on the logit scale and back-transformed so it stays inside [0, 1]; the standard
# error is mapped through the delta method, se_logit = se / (p (1 - p)). Estimates
# are clamped a hair off 0 and 1 only to keep the transform finite.
dom_tbl <- dom_meta |>
  mutate(p = as.numeric(dom_est), se = as.numeric(dom_se)) |>
  mutate(p_c = pmin(pmax(p, 1e-6), 1 - 1e-6),
         gse = se / (p_c * (1 - p_c)),
         lo  = plogis(qlogis(p_c) - 1.96 * gse),
         hi  = plogis(qlogis(p_c) + 1.96 * gse)) |>
  dplyr::select(pvar, plevel, class, p, se, lo, hi)
```

```
walk(cfg$aux, function(v) {
  d <- dom_tbl |> dplyr::filter(pvar == v)
  d |>
    transmute(Level = plevel, Segment = lab_short(class),
              cell = sprintf("%.2f [%.2f, %.2f]", p, lo, hi)) |>
    pivot_wider(names_from = Segment, values_from = cell) |>
    # format = "pipe" + cat(): guaranteed emission under results: asis, and a
    # Pandoc table that wraps instead of overflowing (the kable |> print()
    # pattern is silently swallowed by some render paths).
    knitr::kable(format = "pipe", align = str_c("l", strrep("r", K_star)),
                 caption = paste0("Segment prevalence within ", v,
                                  " (95% JKn intervals; rows sum to one across segments; S1..S
                                  K_star, " named in the segment-label table).")) |>
    cat(sep = "\n")
  cat("\n\n")
  p <- d |>
    mutate(Segment = lab_of(class)) |>
    ggplot(aes(p, plevel, color = Segment)) +
    geom_pointrange(aes(xmin = lo, xmax = hi), size = 0.35, linewidth = 0.4,
                    position = position_dodge(width = 0.6)) +
    scale_color_viridis_d(begin = 0.1, end = 0.8) +
    scale_x_continuous(limits = c(0, NA)) +
    labs(x = "Prevalence of segment membership", y = NULL, color = NULL,
         title = v) +
    theme_lca()
  print(p)
  cat("\n\n\\newpage\n\n")
```

```
})
```

Table 3: Segment prevalence within age_cat (95% JKn intervals; rows sum to one across segments; S1..S5 named in the segment-label table).

Level	S1	S2	S3	S4	S5
18-29	0.36 [0.32, 0.40]	0.09 [0.07, 0.12]	0.21 [0.17, 0.24]	0.09 [0.07, 0.12]	0.25 [0.22, 0.28]
30-44	0.32 [0.28, 0.37]	0.10 [0.07, 0.13]	0.21 [0.17, 0.26]	0.11 [0.08, 0.14]	0.26 [0.21, 0.31]
45-59	0.32 [0.28, 0.37]	0.12 [0.09, 0.16]	0.17 [0.14, 0.22]	0.12 [0.09, 0.16]	0.26 [0.22, 0.30]
60+	0.30 [0.25, 0.36]	0.15 [0.11, 0.20]	0.26 [0.21, 0.31]	0.11 [0.08, 0.15]	0.18 [0.13, 0.24]

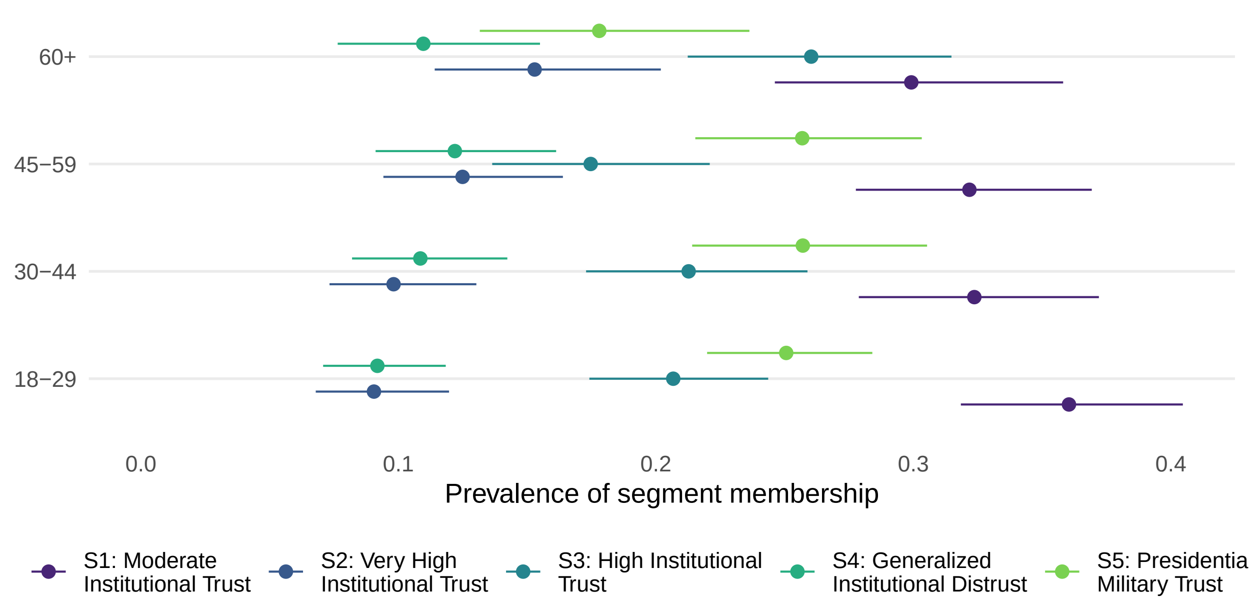


Table 4: Segment prevalence within male (95% JKn intervals; rows sum to one across segments; S1..S5 named in the segment-label table).

Level	S1	S2	S3	S4	S5
Female	0.34 [0.31, 0.38]	0.11 [0.09, 0.14]	0.19 [0.16, 0.22]	0.12 [0.10, 0.14]	0.24 [0.21, 0.27]
Male	0.32 [0.29, 0.35]	0.11 [0.09, 0.13]	0.23 [0.20, 0.26]	0.10 [0.08, 0.12]	0.24 [0.21, 0.28]

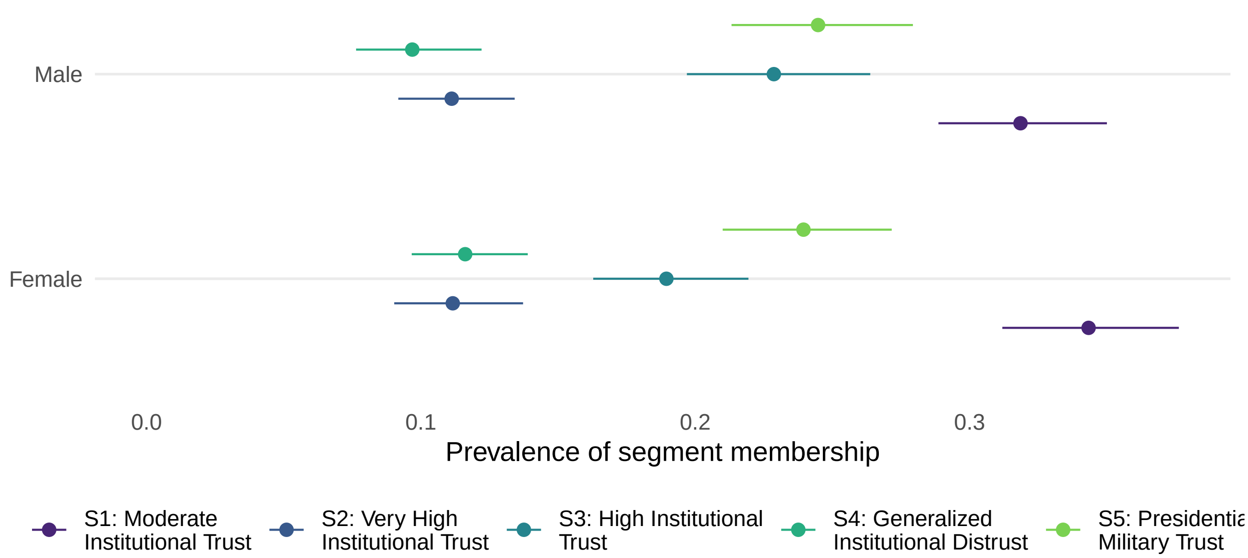


Table 5: Segment prevalence within Edu (95% JKn intervals; rows sum to one across segments; S1..S5 named in the segment-label table).

Level	S1	S2	S3	S4	S5
higher_ed	0.32 [0.27, 0.37]	0.07 [0.04, 0.11]	0.17 [0.13, 0.21]	0.10 [0.07, 0.14]	0.34 [0.29, 0.40]
none	0.11 [0.03, 0.29]	0.41 [0.23, 0.62]	0.15 [0.05, 0.33]	0.21 [0.09, 0.42]	0.13 [0.04, 0.31]
primary	0.33 [0.28, 0.40]	0.17 [0.13, 0.22]	0.22 [0.16, 0.28]	0.09 [0.06, 0.14]	0.19 [0.14, 0.24]
secondary	0.34 [0.31, 0.37]	0.10 [0.08, 0.12]	0.22 [0.20, 0.25]	0.11 [0.09, 0.13]	0.23 [0.20, 0.26]

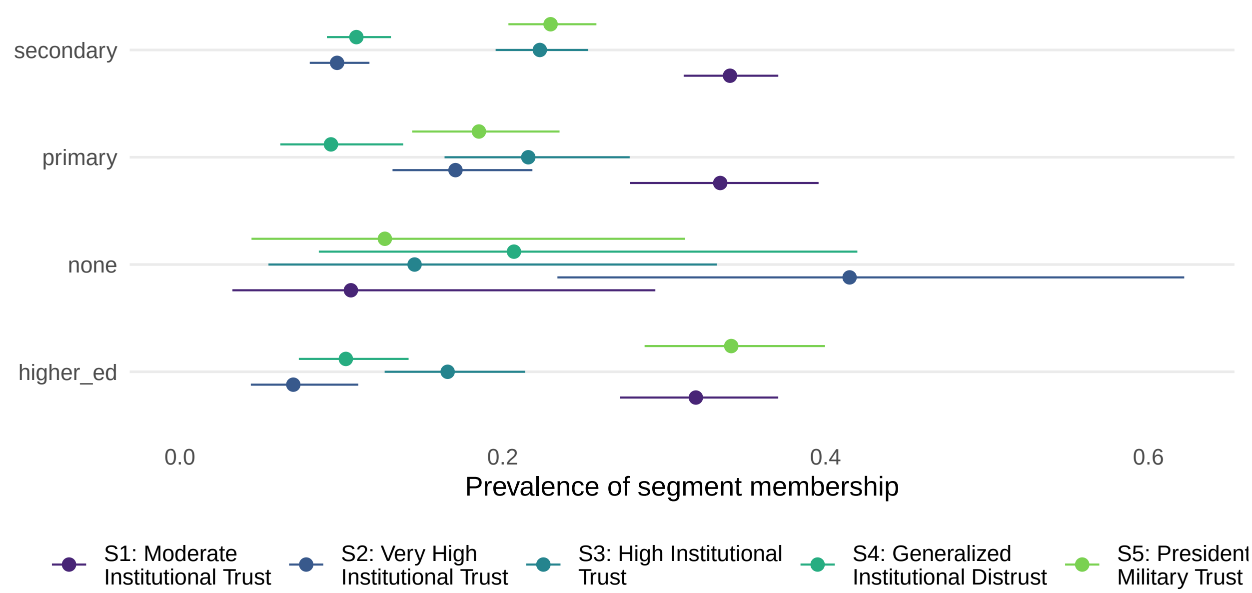


Table 6: Segment prevalence within Urban (95% JKn intervals; rows sum to one across segments; S1..S5 named in the segment-label table).

Level	S1	S2	S3	S4	S5
rural	0.35 [0.30, 0.40]	0.12 [0.09, 0.16]	0.25 [0.20, 0.31]	0.09 [0.06, 0.14]	0.19 [0.15, 0.23]
urban	0.33 [0.30, 0.35]	0.11 [0.09, 0.13]	0.20 [0.17, 0.23]	0.11 [0.09, 0.13]	0.25 [0.23, 0.28]

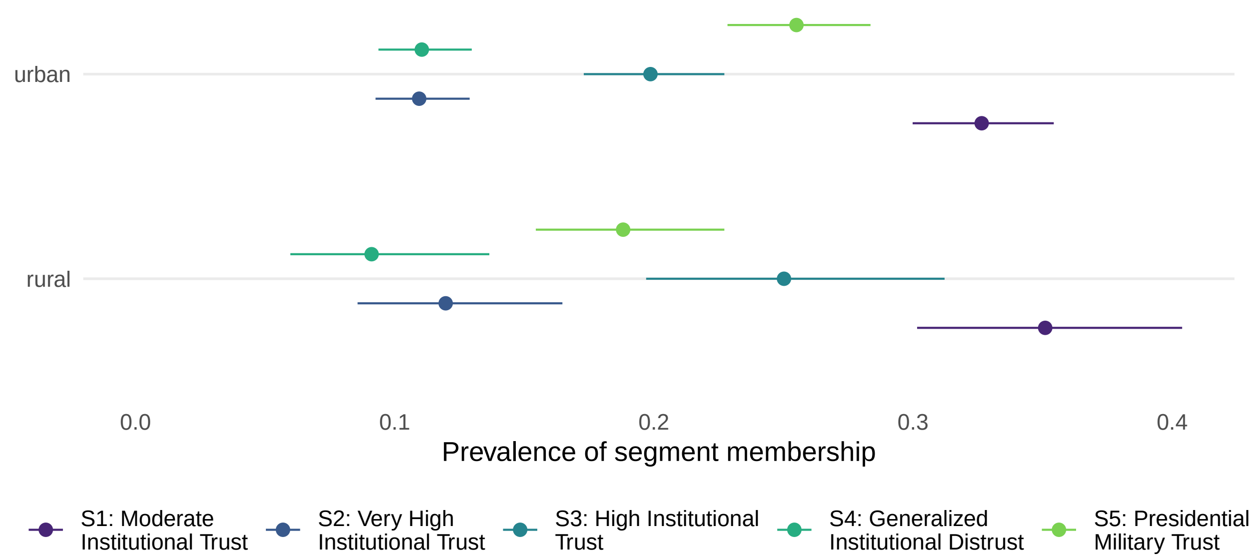


Table 7: Segment prevalence within Employment (95% JKn intervals; rows sum to one across segments; S1..S5 named in the segment-label table).

Level	S1	S2	S3	S4	S5
Employed	0.33 [0.30, 0.36]	0.10 [0.08, 0.12]	0.20 [0.17, 0.23]	0.12 [0.10, 0.14]	0.24 [0.21, 0.27]
house_keeper	0.32 [0.27, 0.37]	0.13 [0.10, 0.17]	0.22 [0.18, 0.27]	0.09 [0.07, 0.12]	0.24 [0.20, 0.28]
Retired	0.34 [0.26, 0.43]	0.17 [0.10, 0.26]	0.19 [0.13, 0.27]	0.10 [0.05, 0.17]	0.20 [0.14, 0.29]
Student	0.38 [0.30, 0.47]	0.09 [0.05, 0.16]	0.25 [0.18, 0.34]	0.07 [0.03, 0.16]	0.21 [0.14, 0.29]
Unemployed	0.32 [0.24, 0.41]	0.09 [0.05, 0.15]	0.19 [0.12, 0.28]	0.09 [0.05, 0.16]	0.31 [0.23, 0.40]

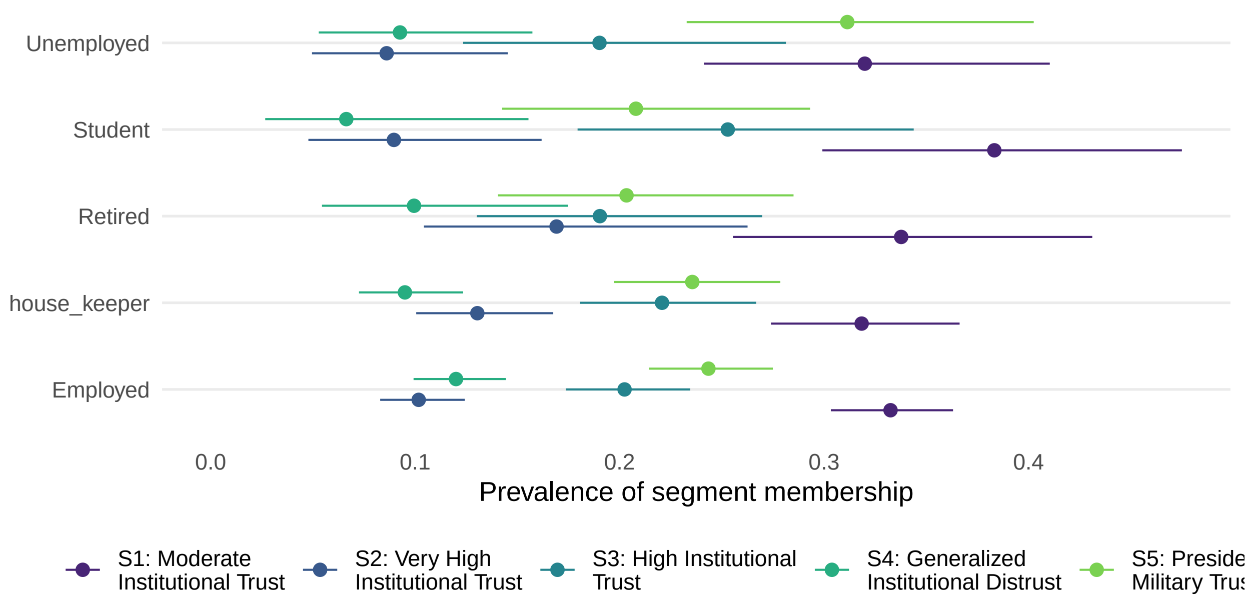


Table 8: Segment prevalence within satis_demo (95% JKn intervals; rows sum to one across segments; S1..S5 named in the segment-label table).

Level	S1	S2	S3	S4	S5
Dissatisfied	0.34 [0.31, 0.39]	0.04 [0.03, 0.06]	0.14 [0.11, 0.17]	0.15 [0.12, 0.18]	0.33 [0.29, 0.37]
Satisfied	0.35 [0.32, 0.38]	0.14 [0.11, 0.17]	0.26 [0.23, 0.30]	0.05 [0.04, 0.07]	0.19 [0.17, 0.22]
Very dissatisfied	0.23 [0.16, 0.33]	0.05 [0.03, 0.11]	0.11 [0.06, 0.19]	0.34 [0.25, 0.43]	0.27 [0.19, 0.37]
Very satisfied	0.21 [0.15, 0.30]	0.33 [0.25, 0.43]	0.22 [0.15, 0.31]	0.05 [0.02, 0.11]	0.18 [0.12, 0.27]

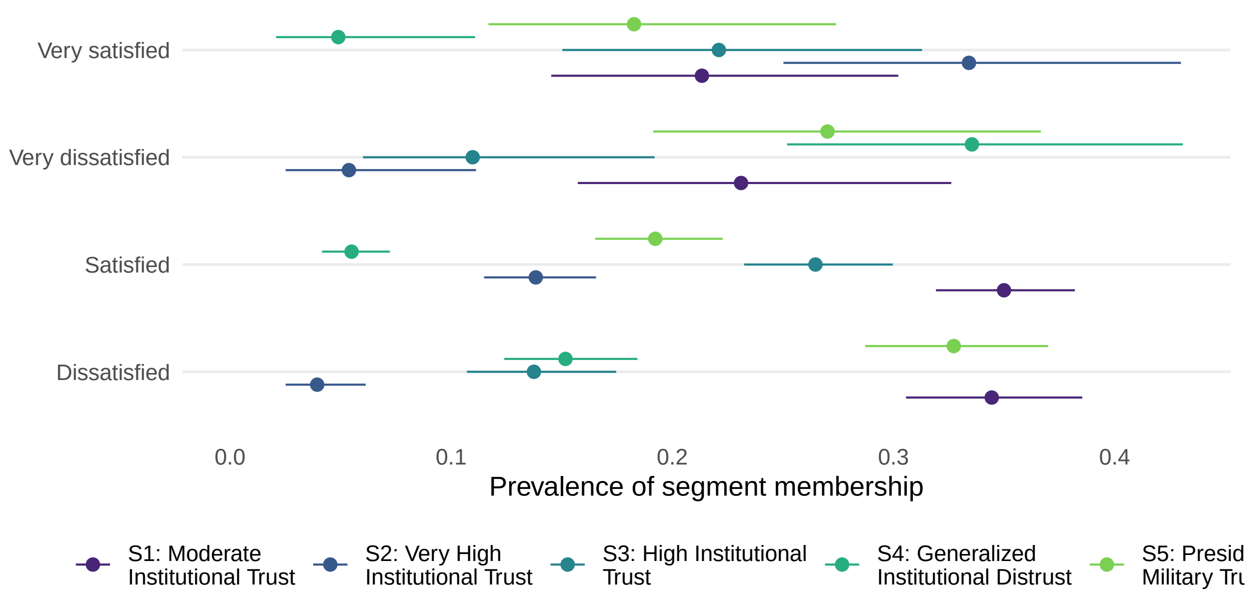
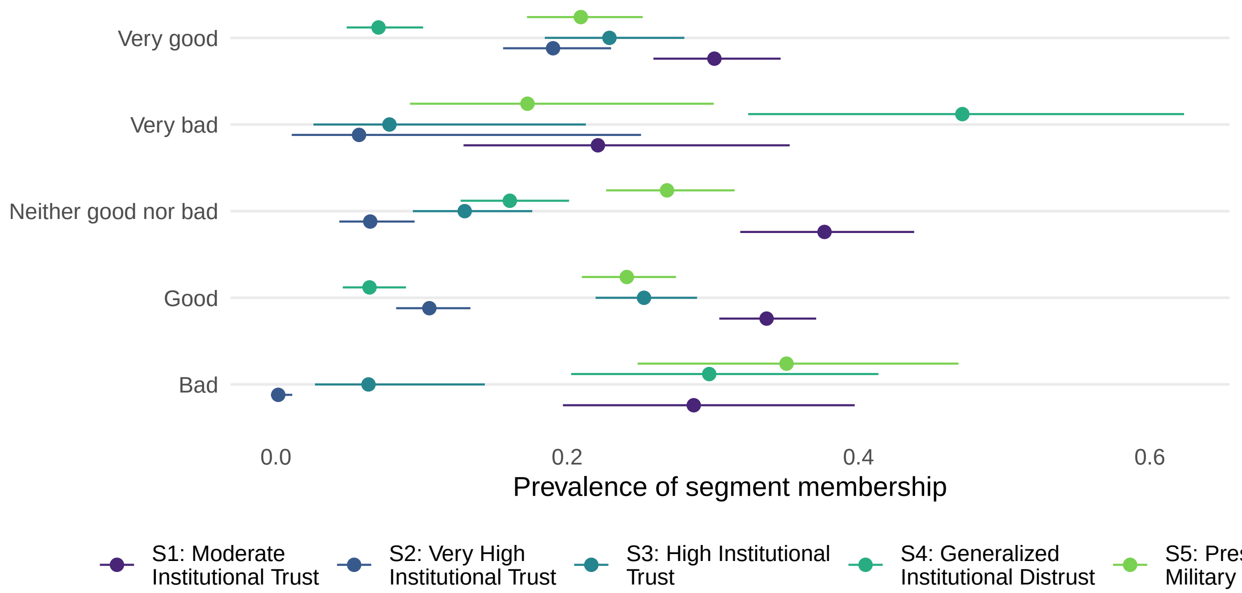


Table 9: Segment prevalence within prez_rating (95% JKn intervals; rows sum to one across segments; S1..S5 named in the segment-label table).

Level	S1	S2	S3	S4	S5
Bad	0.29 [0.20, 0.40]	0.00 [0.00, 0.01]	0.06 [0.03, 0.14]	0.30 [0.20, 0.41]	0.35 [0.25, 0.47]
Good	0.34 [0.30, 0.37]	0.11 [0.08, 0.13]	0.25 [0.22, 0.29]	0.06 [0.05, 0.09]	0.24 [0.21, 0.27]
Neither good nor bad	0.38 [0.32, 0.44]	0.06 [0.04, 0.10]	0.13 [0.09, 0.18]	0.16 [0.13, 0.20]	0.27 [0.23, 0.31]
Very bad	0.22 [0.13, 0.35]	0.06 [0.01, 0.25]	0.08 [0.03, 0.21]	0.47 [0.32, 0.62]	0.17 [0.09, 0.30]
Very good	0.30 [0.26, 0.35]	0.19 [0.16, 0.23]	0.23 [0.18, 0.28]	0.07 [0.05, 0.10]	0.21 [0.17, 0.25]



8 The handoff

Three artifacts: the labelled `.sav` for analysts (numeric `segment` with the LLM/analyst labels as value labels, `NA` below the prediction floor, plus `segment_label` as text and the diagnostics from prediction), and CSVs of the measurement model and the demographic results.

```
seg_labelled <- haven::labelled(
  survey_scored$segment,
  labels = set_names(segment_labels$K, segment_labels$Label),
  label  = "Segment (design-weighted LCA)")
out_sav <- survey_scored |>
  mutate(segment = seg_labelled,
         segment_label = ifelse(is.na(segment), NA_character_,
                                segment_labels$Label[survey_scored$segment]))
haven::write_sav(out_sav, file.path(cfg$out_dir, "survey_with_segments.sav"))

readr::write_csv(ci_tbl |>
                   mutate(segment_label = segment_labels$Label[class]),
                 file.path(cfg$out_dir, "measurement_model_with_CIs.csv"))
readr::write_csv(dom_tbl |>
                   transmute(demographic = pvar, level = plevel,
                             segment = class,
                             segment_label = segment_labels$Label[class],
                             prevalence = p, lo, hi),
                 file.path(cfg$out_dir, "segment_demographics.csv"))
cat("Wrote survey_with_segments.sav (", sum(!is.na(out_sav$segment)), "assigned ),",
    "measurement_model_with_CIs.csv, segment_demographics.csv to", cfg$out_dir, "\n")
```

```
Wrote survey_with_segments.sav ( 1618 assigned ), measurement_model_with_CIs.csv, segment_demog
```